

```
# Import Required Libraries

import pandas as pd

import numpy as np

from sklearn.preprocessing import LabelEncoder
```

```
# Read Dataset

df = pd.read_csv("dataset/loan_data.csv")
```

```
# -----
```

```
# Display Dataset Information
```

```
# -----
```

```
print("Dataset Shape:")
```

```
print(df.shape)
```

```
print("\nDataset Information:")
```

```
print(df.info())
```

```
print("\nMissing Values:")
```

```
print(df.isnull().sum())
```

```
# -----
```

```
# Handle Missing Numerical Values
```

```
# -----
```

```
numerical_columns = [  
    'ApplicantIncome',  
    'CoapplicantIncome',  
    'LoanAmount',  
    'Loan_Amount_Term'  
]  
  
for column in numerical_columns:  
    df[column].fillna(df[column].mean(), inplace=True)  
  
# -----  
# Handle Missing Categorical Values  
# -----  
  
categorical_columns = [  
    'Gender',  
    'Married',  
    'Dependents',  
    'Education',  
    'Self_Employed',  
    'Credit_History',  
    'Property_Area',  
    'Loan_Status'  
]
```

```
for column in categorical_columns:

    df[column].fillna(df[column].mode()[0], inplace=True)

# -----

# Encode Categorical Variables

# -----

label_encoder = LabelEncoder()

for column in categorical_columns:

    df[column] = label_encoder.fit_transform(df[column])

# -----

# Verify Missing Values

# -----

print("\nMissing Values After Preprocessing:")
print(df.isnull().sum())

# -----

# Display Cleaned Dataset

# -----

print("\nFirst Five Rows:")
print(df.head())
```

```
print("\nData Types:")
```

```
print(df.dtypes)
```

```
print("\nDataset Shape After Preprocessing:")
```

```
print(df.shape)
```